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Microbiome-based prediction of allogeneic hematopoietic stem cell transplantation outcome

Oshrit Shtossel^{1†}, Adi Eshel^{2†}, Shalev Fried^{3,4}, Mika Geva^{3,4}, Ivetta Danylesko^{3,4}, Ronit Yerushalmi^{3,4}, Noga Shem-Tov^{3,4}, Joshua A. Fein⁵, Marco Fabbrini⁶, Avichai Shimoni^{3,4}, Sondra Turjeman², Yoram Louzoun¹, Arnon Nagler^{3,4†}, Omry Koren^{2†} and Roni Shouval^{3,4,5,7*†}

Abstract

Background Allogeneic hematopoietic stem cell transplantation (HSCT) is potentially curative for hematologic malignancies but is frequently complicated by relapse and immune-mediated complications, such as graft-versus-host disease (GVHD). Emerging evidence suggests a role for the intestinal and oral microbiome in modulating HSCT outcomes, yet predictive models incorporating microbiome data remain limited.

Methods We applied the RATIO (suRvival Analysis lefT barrler IOss) model to longitudinal stool and saliva microbiome data from 204 adult HSCT recipients to predict the timing of seven outcomes: overall survival (OS), non-relapse mortality (NRM), relapse, acute GVHD (grades II–IV and III–IV), chronic GVHD, and oral chronic GVHD. A total of 514 stool and 1291 saliva samples were collected over 70 weeks post-HSCT. Model performance was evaluated using the concordance index (CI) and Spearman correlation coefficient (SCC), with SHAP (SHapley Additive exPlanations) analysis used for model interpretability.

Results Oral and stool microbial dysbiosis peaked within the first 2 weeks post-HSCT, followed by partial recovery. Using the RATIO model, we found that microbiome features from early time points (weeks 1–2) were most predictive of short-term complications such as acute GVHD, while later samples (weeks 36–70) were more informative for long-term outcomes, including overall survival. RATIO outperformed traditional survival models (Cox and Random Survival Forest) across most outcomes (median CI > 0.65), with stool microbiota showing greater predictive power than saliva. SHAP analysis identified specific stool genera, including *Collinsella* and *Eggerthella*, associated with shorter time to various complications. External validation using a pediatric GVHD cohort confirmed the model's generalizability and reproducibility. External validation using a pediatric HSCT cohort ($n = 90$) confirmed the reproducibility and generalizability of these microbiome-based predictions.

Conclusions Microbiome profiling of stool and saliva samples offers robust, time-sensitive prediction of post-HSCT complications. The RATIO model enables interpretable, time-to-event prediction across multiple outcomes and may inform microbiome-guided interventions to improve transplant success.

[†]Oshrit Shtossel, Adi Eshel, Arnon Nagler, Omry Koren and Roni Shouval have contributed equally.

*Correspondence:

Roni Shouval
shouval@gmail.com

Full list of author information is available at the end of the article



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Keywords Hematopoietic stem cell transplantation, Stool microbiome, Saliva microbiome, Time-to-event analysis, Machine learning, Graft-versus-host disease, RATIO model

Background

Allogeneic hematopoietic stem cell transplantation (HSCT, see Additional file 1: Table S1 for Acronym Table) is a curative treatment for both malignant and non-malignant hematological disorders [1]. Patients undergoing HSCT receive conditioning regimens comprising chemotherapy and/or radiation to reduce disease burden and mitigate the risk of graft rejection. Moreover, the procedure entails intensive immunosuppression aimed at preventing graft-versus-host disease (GVHD). Despite recent improvements in HSCT outcomes [2], infections [3], GVHD [4], disease recurrence, and other transplantation-related complications continue to pose significant challenges, with approximately 50% of patients experiencing relapse or non-relapse mortality (NRM) [5].

The microbiota—a population of commensal organisms colonizing the host—is integral to numerous host functions, including digestion, immunity, endocrine regulation, and neural processes. It is also increasingly recognized for its influence on disease outcomes and treatment responses [6–10]. In the context of HSCT, the host microbiota, across body sites, is vastly altered and dynamic pre-transplantation and also following HSCT due to the treatment regimen for the underlying disorder, preparatory treatment, like with broad-spectrum antibiotics and intensive chemotherapy, prophylaxis regimens, and more [11–15]. Furthermore, there is mounting evidence that HSCT complications and mortality can be related to the host microbiome [16–21]. For instance, a dysbiotic fecal microbiome is associated with increased NRM and acute GVHD (AGVHD) [22, 23]. At the taxonomic level, expansion of the genus *Enterococcus* has been implicated in elevated GVHD risk, while the presence of specific bacterial taxa, including *Eubacterium limosum*, correlates with reduced relapse risk [23, 24]. The oral microbiota is also implicated in HSCT complications including oral mucositis and GVHD [11, 25].

Despite the growing body of literature on the pivotal role of the microbiome in HSCT, findings are rarely translated into actionable information that would support precision treatment. Prediction models integrating microbiome are challenged by the high-dimensional nature of the microbiome (i.e., the number of taxa > number of patients). Furthermore, methods such as Kaplan–Meier estimator and/or Cox survival model, used for survival estimation and modeling, are not appropriate for competing risk scenarios, such as GVHD prediction [26–29] which should take into account time to

event as well as the alternative events of death or relapse [30, 31].

Several studies have addressed the issue of competing events by modifying their objective to estimate risks [32–36]. An alternative approach is the utilization of a semi-competing risks framework [37, 38]. These approaches can be categorized into three main groups: copula-based models that employ joint probability models [39, 40], models rooted in causal inference [41, 42], and models based on the illness-death multi-state model [43, 44]. However, these models primarily focus on returning a risk estimation rather than considering the time to event (TTE), and they primarily focus on a single complication. To address these limitations, we have developed a novel model that enables clinicians to make informed, data-driven decisions for their patients.

We hypothesize that integrating patients' clinical data with microbiome profiles at multiple time points can improve the predictive accuracy for HSCT-associated complications. This approach may enable more precise preventive interventions while illuminating the biological mechanisms driving these complications.

Here, we apply RATIO (suRvival Analysis left barrier loss) [45] on microbial data (stool and saliva) to predict the time to HSCT complications. RATIO effectively incorporates information from highly censored data and is compatible with a wide range of models, from simple linear regressions to convolutional neural networks (CNNs). To our knowledge, this is the first study to utilize RATIO on a large cohort, integrating microbial data from two distinct body sites (saliva and stool) to predict TTE outcomes for HSCT complications and validate the model's microbial associations.

Methods

Cohort description and sample collection

This was a **single-center prospective study** conducted at the Chaim Sheba Medical Center (Tel-Hashomer, Israel) between 2016 and 2018, complemented by a pediatrics external validation cohort [64] (see below). Adult patients undergoing allogeneic HSCT were enrolled upon hospital admission. There were no restrictions based on transplant indication, donor type, or conditioning regimen. Clinical data and metadata were managed using a secure REDCap database [46].

The primary cohort included 204 patients, from whom 514 stool samples ($n=154$ patients) and 1291 saliva samples ($n=201$ patients) were collected over a 70-week

period post-transplant. Sample collection began approximately 7 days prior to stem cell infusion (day 0) and continued weekly during the inpatient phase and monthly after discharge, up to 485 days. Sampling frequency varied according to hospitalization status and patient compliance. Stool and saliva microbiota were analyzed independently and in combination.

Saliva was collected by passive drooling (2–5 mL), aliquoted into 1 mL cryovials, and immediately frozen in liquid nitrogen. Stool (~1 tablespoon) was self-collected, similarly aliquoted, and processed identically. All samples were stored at -80°C without preservative medium.

Clinical data and outcomes

Clinical variables included age, sex, Karnofsky Performance Status (KPS < 90 or ≥ 90), body mass index (BMI), albumin, disease risk, donor type, conditioning intensity, GVHD prophylaxis, comorbidity burden (HCT-CI < 3 or ≥ 3), and renal function (eGFR < 60 or ≥ 60 mL/min/1.73 m²). Categorical variables were one-hot encoded; continuous variables were standardized using z-scores. All features were collected before conditioning and used in standalone or integrated models.

Conditioning intensity was classified using Transplantation Conditioning Intensity (TCI) criteria [94]. GVHD prophylaxis included cyclosporine A (CsA) with methotrexate (MTX) or mycophenolate mofetil (MMF), with anti-thymocyte globulin (ATG) administered for unrelated or mismatched donors. Haploidentical transplants received post-transplant cyclophosphamide. Antimicrobial prophylaxis included trimethoprim/sulfamethoxazole, fluconazole, valganciclovir, and (when applicable) ganciclovir. Empiric treatment for neutropenic fever included piperacillin-tazobactam and amikacin; meropenem was reserved for hemodynamic instability. Neutrophil engraftment was defined as three consecutive days with ANC $> 0.5 \times 10^9/\text{L}$.

The study evaluated seven post-HSCT outcomes: overall survival (OS), non-relapse mortality (NRM), relapse, acute GVHD (grades II–IV and III–IV), chronic GVHD, and oral chronic GVHD. OS was defined from the date of stem cell infusion until death or last follow-up. Acute GVHD was graded per IBMTR criteria [47] and chronic GVHD per NIH consensus guidelines [48]. Relapse and NRM were treated as competing events where appropriate. Full definitions and event classification are provided in Additional File 1: Table S3.

DNA extraction

DNA extraction was performed using the PureLink Microbiome DNA Purification Kit, as described by the manufacturer protocol for stool and saliva samples (Invitrogen, Carlsbad, California, USA) following a 2-min beadbeating

step. Bacterial 16S rRNA gene sequences were PCR-amplified from each sample using the barcoded 515F-806R primers for the V4 hypervariable region of the gene as previously described [11, 49, 50]. The PCR reactions consisted of 25 μL PrimeStar max PCR mix (Takara, Mountain View, California, USA), 2 μL of each primer (10 μM), 4 μL DNA in a total volume of 50 μL . The PCR cycle conditions were: 30 cycles of denaturation (98°C for 10 s), annealing (55°C for 5 s) and extension (72°C for 20 s), with final elongation at 72°C (for 1 min). Amplicons were purified (AMPure, Beckman Coulter, Brea, California, USA), quantified by PicoGreen assay (Molecular Probes, Eugene, OR), and pooled at equal concentrations (30 ng/ μL). Finally, pooled samples were purified using 2% E-Gel (Invitrogen, Carlsbad, CA) and sequenced using the Illumina MiSeq platform at the Genomic Center of the Azrieli Faculty of Medicine, Bar-Ilan University, Israel.

Sequencing and bioinformatic analysis

Data analysis was performed using Quantitative Insights Into Microbial Ecology software (QIIME2) [51]. Paired-end sequences were demultiplexed by per-sample barcodes and error-corrected by Divisive Amplicon Denoising Algorithm (DADA2) [52]. Final feature sequences were aligned against the Greengenes database [53] with 99% confidence for taxonomic annotation. The Shannon diversity index, the alpha diversity parameter indicating microbial richness, was calculated on a rarefied table of > 7000 reads per sample after filtration of sequences of mitochondria and chloroplasts [11].

Preprocessing microbial data

All the 16S rRNA gene sequences of each dataset were processed using the MIPMLP pipeline [54]. The preprocessing of MIPMLP includes four stages: merging similar features based on taxonomy, scaling the distribution, standardization to z scores, and dimension reduction. We merged features at the genus level by averaging all ASVs belonging to the same genus prior to modelling. More formally, assuming that s_{ij} denotes the count of ASV i in sample j , we define $x_{kj} = \text{average}(s_{ij})$ where the average is taken over all ASVs associated with taxon k . To address compositionality, we performed log normalization, inspired by the approach in coda4microbiome [92], followed by z-scoring per sample. Specifically, we compute:

$$z_{kj} = \frac{\log(x_{kj} + \varepsilon) - \text{average}_k(\log(x_{kj} + \varepsilon))}{\text{std}_k(\log(x_{kj} + \varepsilon))}$$

where the average and standard deviation are computed across all taxa k in sample j and epsilon is set to 0.1. All subsequent analyses were performed on the resulting z_{kj} values. No dimensionality reduction was applied at this

stage. For further details, see Additional file 1: Table S4 and Additional file 1: Fig. S1.

Experimental design

To predict time-to-event outcomes for HSCT complications, we implemented three survival models: Cox proportional hazards (COX) [55], Random Survival Forest (RSF) [56, 57], and RATIO [45]. Each model was applied using two input variations: a microbiome-based approach incorporating microbial diversity metrics and features, and a combined approach integrating both microbial and clinical features. Training utilized a Leave-One-Person-Out (LOPO) strategy due to the longitudinal nature of the data, with external validation performed using an independent GVHD cohort. Feature selection, hyperparameter tuning, and evaluation metrics—including Spearman correlation coefficient (SCC) and concordance index (CI)—were used to optimize model performance. Time-point contribution and taxa influence were assessed via iterative microbial feature omission and SHAP analysis, respectively. Detailed methodologies, including hyperparameter optimization and validation specifics, are provided in the Additional file 1 (Tables S5–S8).

External validation cohort (Masetti et al. [64])

We validated the RATIO model using a publicly available dataset from Masetti et al. [64], comprising 90 pediatric allogeneic HSCT recipients from five Italian centers. Stool samples (*n* = 180) were collected pre-transplant and at neutrophil engraftment and processed using 16S rRNA sequencing (V4 region) with Illumina MiSeq. Sequencing data were analyzed using QIIME2 and DADA2, with genus-level features derived via the MIPMLP pipeline following log normalization and z-scoring.

The RATIO model was applied to predict overall survival and acute GVHD based on microbiome features alone and in combination with key clinical variables. SHAP-based interpretation and model transferability were assessed using the same pipeline as in the primary cohort. Full processing and analysis details are provided in Additional file 1.

Results

Population characteristics and outcomes

A total of 204 patients with a median age of 58 years (interquartile range (IQR) 46 to 67) were included (Table 1). Acute myeloid leukemia (AML) was the leading diagnosis (48%). Most donors were human leukocyte antigen (HLA)-matched unrelated (52%) and HLA-matched related (36%). Intermediate intensity conditioning (58%) and ATG (59%) were used in the majority of patients. With a median follow-up of 26.3 months (IQR 23.7–31.1), the 2-year overall survival (OS) probability

Table 1 Population characteristics

Characteristic	N = 204 ¹
Age	58 (46, 67)
Sex	
Female	70 (34%)
Male	134 (66%)
Karnofsky Performance status	
< 90	42 (21%)
≥ 90	157 (79%)
Unknown	5
HCT-CI	
0	40 (20%)
1–2	79 (39%)
3+	85 (42%)
Diagnosis	
Acute myeloid leukemia	98 (48%)
Lymphoma	34 (17%)
Myelodysplastic syndrome	32 (16%)
Myeloproliferative neoplasms	17 (8%)
Acute lymphoblastic leukemia	15 (7%)
Other	8 (4%)
Donor type	
HLA-matched unrelated donor	106 (52%)
HLA-matched related donor	74 (36%)
Haploidentical donor	13 (6%)
HLA-mismatched unrelated donor	11 (5%)
Conditioning intensity	
Low intensity	68 (33%)
Intermediate intensity	118 (58%)
High intensity	18 (8.8%)
GVHD prophylaxis with methotrexate	149 (73%)
In vivo T-cell depletion	
None	71 (35%)
ATG	121 (59%)
ATG and PTCy	10 (4.9%)
PTCy	2 (1.0%)

¹ Median (IQR); *n* (%)

was 64% (95% CI 58–71%). The cumulative incidence of NRM and relapse at 2 years was 11% (7–16%) and 37% (30–43%), respectively. Day 100 cumulative incidence of acute GVHD grade II–IV was 25% (19–31%), and it was 12% (8–17%) for grade III–IV acute GVHD, respectively. Two years after HSCT, the cumulative incidence of chronic (CGVHD) and oral CGVHD was 34% (27–40%) and 13% (9–18%), respectively.

Stool and saliva samples

A total of 514 stool and 1291 saliva samples were obtained from 154 and 201 allogeneic HSCT recipients,

respectively. Samples were collected at different time points, starting from 7 days before the transplantation and up to 485 days post-HSCT. Most samples (62%) were collected during the first 100 post-transplant, with an average of 2.95 stool samples and 5.27 saliva samples per patient, respectively (Fig. S2). The most common HSCT indication was acute leukemia, with 50.3% for the stool cohort and 40.1% for the saliva. Median follow-up was estimated using the reverse Kaplan–Meier method with the *prodlm* package and reported with interquartile range [58]; it was 26.3 months. Samples were collected weekly at first (first 4 weeks post-transplant) and later, less frequently, which resulted in a median of 3 samples per patient, with an IQR of 2 for the stool cohort and 6 for the saliva.

Stool and saliva microbiome characteristics

Microbial characterization revealed significant changes in diversity and bacterial dominance following transplantation. A notable reduction in microbial richness, measured by Shannon's diversity index (alpha diversity), was observed in both stool and saliva samples during the first 20 days post-transplantation, followed by a moderate recovery (Fig. 1A, B). A statistically significant Spearman correlation of $r=0.391$, $p<0.0001$ was measured between the stool and saliva microbial richness. Bacterial domination, defined as ASVs with a relative abundance exceeding 0.3, was detected in up to 60% of stool and saliva samples starting at day 14 and day 7, respectively. The genus *Enterococcus* and unclassified members of Enterobacteriaceae emerged as the most prevalent over-dominant genera in stool samples, whereas *Veillonella* and *Streptococcus* were predominant in saliva samples (Fig. 1C, D).

Microbiome predicts HSCT outcomes

We hypothesized that the oral and stool microbiomes independently serve as informative biomarkers for predicting patient outcomes following HSCT. To test this, we examined seven key HSCT-related complications: AGVHD (II–IV), AGVHD (III–IV), CGVHD, oral CGVHD, relapse, NRM, and OS. For each outcome, we aimed to predict the TTE using the microbiome compositions from the collected stool and saliva samples. Accurate TTE estimation is imperative for understanding the progression of these complications and for developing timely interventions. However, censoring and competing events, such as relapse, NRM, can bias TTE estimates if not properly accounted for. To address these challenges, we utilized statistical approaches (the (Additional file 1: Machine Learning Nomenclature and Additional file 1: Schematic Fig. S3) designed to handle censoring and competing risks, ensuring robust and reliable predictions of patient outcomes.

We applied Survival Analysis Prediction (for summary of existing methods see Table S2) [59–61] approaches to predict the TTE of the seven transplant complications based solely on the preprocessed microbial data (see the “Methods” section). Given the established relationship between Shannon's diversity index and HSCT outcomes, this index was included as an additional feature in all models. Pre-feature selection was performed on the training data (see the “Methods” section), and three models—COX [55], Random Survival Model (RSF) [56, 57], and RATIO [45]—were applied to the selected features. Model performance was evaluated using the Spearman Correlation Coefficient (SCC) and concordance index (CI) on the test set, which consisted solely of uncensored samples.

The RATIO model demonstrated strong predictive performance for TTEs in both stool and saliva cohorts (CI>0.55; Fig. 2A, B). Similar results were observed when considering SCCs (Fig. S4). Notably, the stool microbiome significantly emerged as a more robust predictor than saliva, with an average CI of 0.656 compared to 0.613 for saliva (p -value=0.02). Nonetheless, the saliva microbiome also showed utility as a biomarker for predicting complications. Among the outcomes, oral CGVHD was the most accurately predicted in both cohorts, whereas NRM and relapse were the least accurately predicted.

Clinical data adds modest value to microbiome-based predictions

Beyond the microbiome's independent predictive capability demonstrated by the RATIO model, we examined the impact of integrating patients' clinical data into a combined model. This approach involved first applying the microbiome-only RATIO model and then incorporating clinical data to create a combined RATIO model. While the inclusion of clinical data led to improvements in both the CI (Fig. 2C, D) and SCC (Fig. S5), the enhancement was modest compared to using microbiome data alone. These findings suggest that the microbiome data captures the majority of predictive information for HSCT complications (Additional file 1: Fig. S6 and Tables S9–S10). Nevertheless, integrating traditional clinical information can provide incremental benefits to the model's predictive performance.

Additionally, we conducted F -tests to formally assess whether adding clinical features to the microbial-only model, and conversely, adding microbial features to the clinical-only model, led to a significant improvement in predictive performance (F -test results are shown in Additional file 1: Tables S9–S10). The results show that: In 13 out of 14 cases (stool and saliva), adding microbial features beyond clinical features led to a statistically

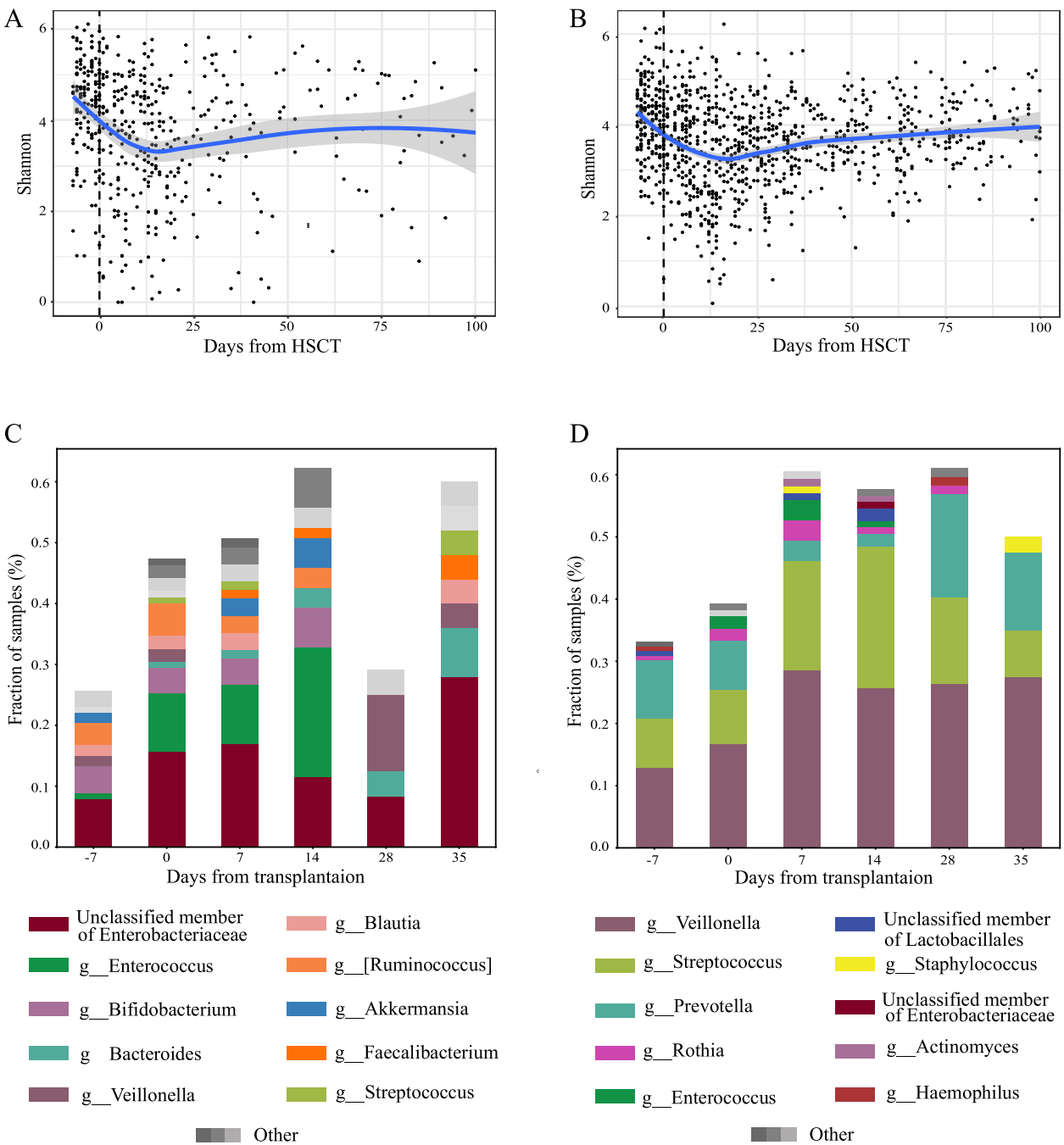


Fig. 1 Stool and saliva microbiota dysbiosis characteristics along HSCT. **A–B** Changes in Shannon index over time for HSCT patients' stool samples (**A**) and saliva samples (**B**). Day zero (dashed black line) represents the day of HSCT, the black dots represent each sample's Shannon index value, and the blue lines represent the moving average for the Shannon index values. **C–D** The fraction of samples represents bacteria domination (ASVs with relative abundance > 0.3) of different genera along the days before and after HSCT. The x-axis represents the days from HSCT and the y-axis represents the percentage of samples with bacteria domination in stool (**C**) and saliva (**D**); colors represent the 10 most abundant taxa at the genus level, while other genera are colored with gray

significant improvement in the SCC. The CI was also improved on average by adding the microbiome data. In 9 out of 14 cases (stool and saliva), the microbial

contribution beyond the clinical features was greater than the clinical contribution beyond the microbial features (see Additional file 1: Fig. S6).

In summary, while the majority of predictive information for HSCT complications resides in the microbiome data, incorporating traditional clinical information into the model can further enhance its predictive ability.

Role of sample timing in microbiome-based HSCT predictions

To determine the impact of different time points in identifying relevant microbiome biomarkers, we constructed prediction models that systematically excluded stool or saliva samples from a specific week using the leave-one-out jackknife method [62]. The percentage change in performance between the model without the excluded time point and the original model utilizing all samples was calculated (refer to methods for details). Our analysis indicated that samples collected one to 2 weeks post-HSCT were most influential for predicting the majority of complications in both stool and saliva cohorts. For long-term outcomes, such as OS, samples collected at later stages (36 weeks to 70 weeks post-HSCT) contributed more to model performance. In contrast, outcomes like NRM and relapse showed limited sensitivity to the timing of microbial sampling (Fig. 2E, F).

Microbiome involvement differs according to the complication post-HSCT

In addition to its enhanced performance, RATIO offers the potential for identifying taxa that are closely associated with a specific condition. We used SHAP (a game theoretic approach to explain the output of machine learning models [63]), to estimate which taxa contributed most to the model and in which direction. The SHAP model matches a value of the effect of the feature values with the model prediction. We further calculated the sum of SHAP values and the input values for each feature to obtain a directed score of the overall contribution.

Figure 3 illustrates the correlation between different genera of stool (Fig. 3A) and saliva (Fig. 3B) with the time to occurrence of each of the seven investigated

events. Overall, the correlation between saliva and time to event exhibited a well-clustered pattern, except for the events OS and NRM, indicating similar taxonomic contributions across different genera. However, in the stool cohort, the clustering was less pronounced.

Specifically, a high relative abundance of stool taxa *Collinsella*, *Eggerthella*, *Alistipes*, *Parabacteroides*, *Bacteroides*, and *Paraprevotella* showed a negative correlation, with a shorter time to all seven events (Fig. 3A). On the other hand, stool taxa *Corynebacterium* and an unclassified member of Actinomycetaceae displayed a positive correlation with the different events. Regarding saliva, the bacteria correlation with the events of oral CGVHD, CGVHD, AGVHD (III–IV), AGVHD (II–IV), had a contrasting relationship with the events of OS and NRM. Specifically, the taxa associated with various graft-versus-host disease events demonstrated an inverse impact on the occurrences of OS and NRM.

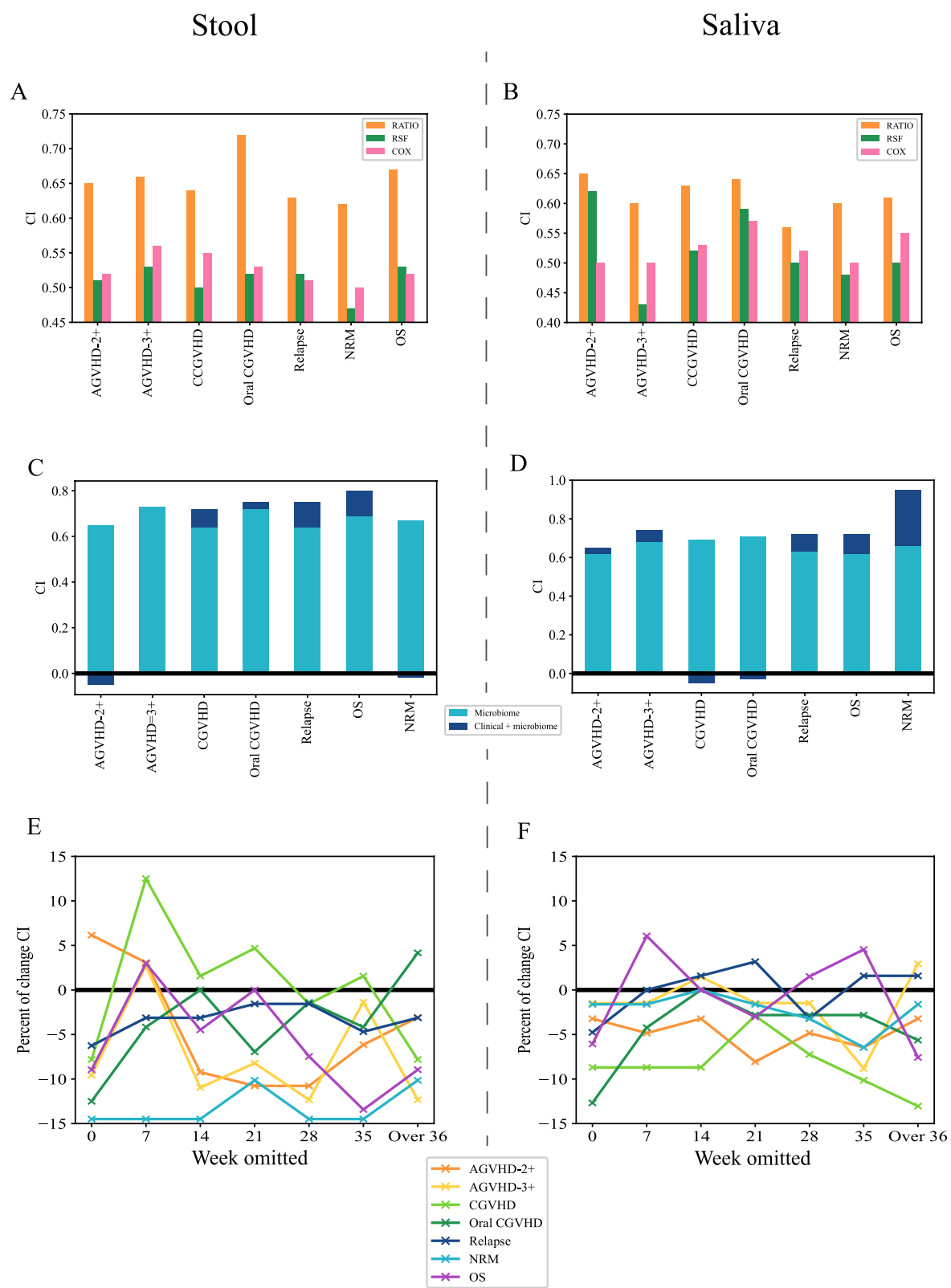
Validation of RATIO's generalizability across cohorts

To ensure that microbial associations identified by RATIO are not limited to the current study, we analyzed an additional GVHD study from Masetti et al. [64] as a validation cohort. RATIO was applied to the microbiome data to predict the time to either AGVHD or OS, achieving prediction performance comparable to that observed in our study (CI: 0.65 and SCC: above 0.45 in LOPO; Fig. 4A). Incorporating clinical data such as age, sex, donor stem cell source, and conditioning intensity led to a slight improvement in time-to-event (TTE) predictions (Fig. 4B, C).

Next, we examined whether the microbial associations detected by RATIO were consistent across the two cohorts by analyzing the direction of SHAP values from both analyses. We identified 12 consistent associations for AGVHD predictions and 11 for OS predictions (Fig. 4D, E). For instance, *Collinsella*, *Eggerthella*, *Methanobrevibacter*, *Alistipes*, *Bifidobacterium*, *Parabacteroides*, and *Bacteroides*. Furthermore, most of the

(See figure on next page.)

Fig. 2 Microbiome predicts HSCT complications. **A–B** Different models (RATIO, RSF, and COX) CI scores based on the stool microbiome (**A**) and the saliva microbiome (**B**). The orange bar represents the RATIO score, the green bar represents the COX score, and the pink bar represents the RSF score. The results are reported on a leave-one-person-out learning approach (LOPO). The same results with the SCC metric can be found in Fig. S4. **C–D** Clinical data features further improve HSCT complication prediction over the microbiome data in the stool cohort (**C**) and the saliva cohort (**D**). The light bars represent the RATIO microbiome-based CI score, while the dark bars represent the further improvement of the clinical data. The black line represents the zero line of no improvement; when the dark bar is positive, the clinical data contribution is positive, whereas when the dark bar is negative, the clinical data contribution is negative. The results are reported on a leave-one-person-out learning approach (LOPO). Parallel results with SCC can be found in Fig. S5. **E–F** Analysis of the most critical time of the microbiome sampling to the prediction quality in the stool cohort (**E**) and the saliva cohort (**F**). The x-axis represents weeks intervals before and after HSCT. The y-axis represents the change in CI score while omitting the data from a certain week. If the change is positive, the time point is not critical to the learning, and omitting it further improves the learning. If the change is negative, the time point is critical to the learning, and omitting it harms the learning performance. The results are reported on a leave-one-person-out learning approach (LOPO)



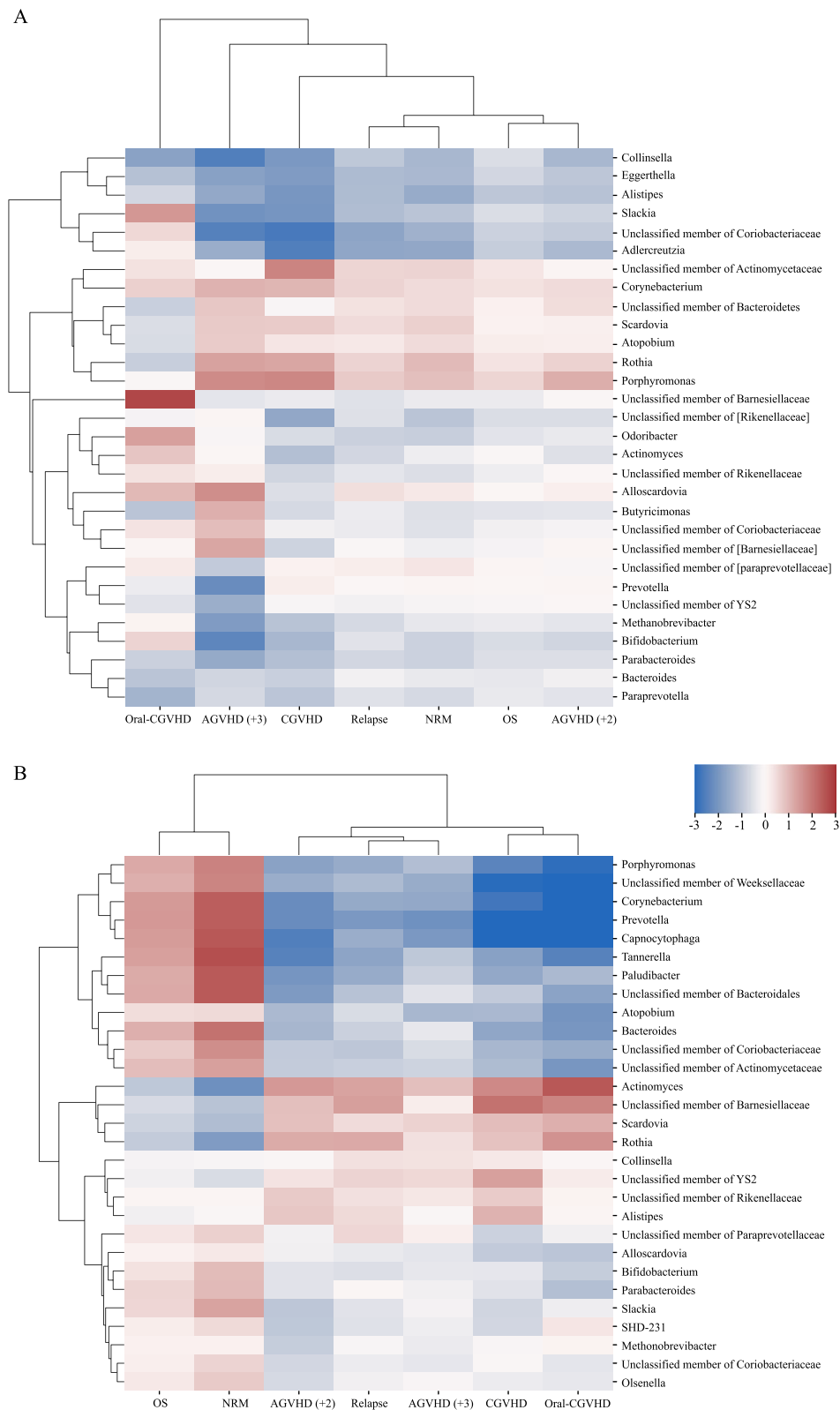


Fig. 3 Microbiome involvement differs according to the complication post-HSCT. **A–B** Heatmaps of RATIO the directed contribution score based on the SHAP values over the different HSCT complications and outcomes in stool (**A**) and saliva (**B**). While blue represents a negative correlation, meaning higher relative abundance correlated with a shorter time to the event. Red represents a positive correlation, meaning higher relative abundance correlated with a longer time to the event

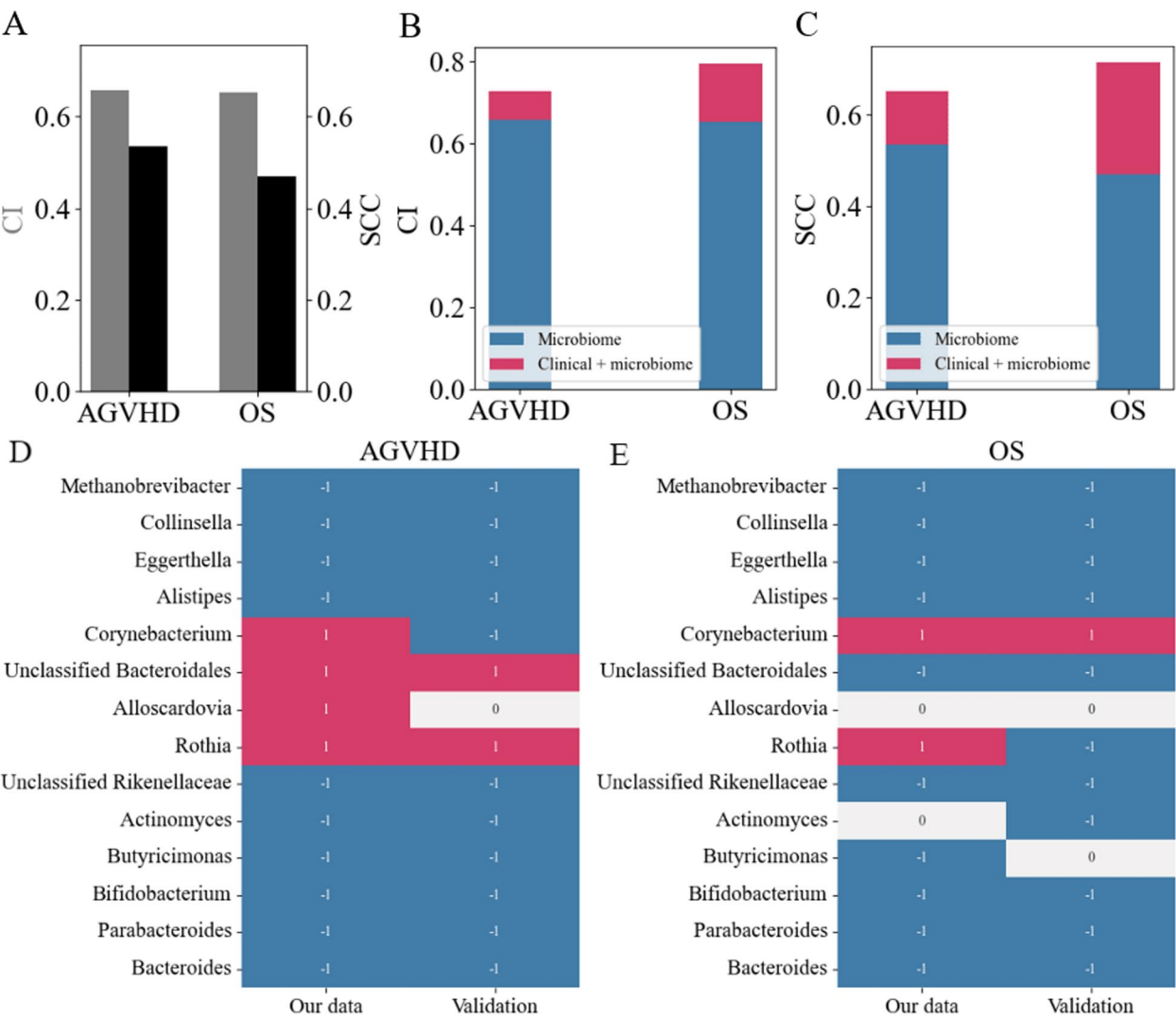


Fig. 4 Massetti et al. stool validation cohort. **(A)** Microbiome predicts HSCT complications. RATIO’s CI (left gray) and SCC scores (black right) based on the stool microbiome. The results are reported on a leave-one-person-out learning approach (LOPO). **B–C** Clinical data features further improve HSCT complication prediction over the microbiome data in the validation cohort. The blue bars represent RATIO microbiome-based CI score **(B)** or SCC scores **(C)**, while the red bars represent the further improvement of the clinical data. **D–E** Heatmaps of taxa used by the RATIO model for HSCT predictions, including AGVHD **(D)** and OS **(E)**, in our study (“Our data”) and the validation study (“Validation”). The colors represent the direction of the SHAP effect on the RATIO model for each genus. Blue indicates a negative effect, corresponding to shorter time-to-event predictions, while red indicates a positive effect, corresponding to longer time-to-event predictions. White indicates that the specific genus was not detected in the respective study

multivariate associations identified by RATIO in the validation cohort are consistent with the univariate associations reported in the original paper (Fig. S7). We further validated that RATIO can be used for cross-dataset prediction capability by training RATIO on our dataset and testing on the Masetti et al. data, achieving a CI of 0.55 and 0.57 for predicting outcomes (AGVHD and OS, respectively) in the validation cohort.

Discussion
This single-center observational study provides valuable insights into the potential of the stool and saliva microbiome as essential biomarkers for predicting the timing of various events following HSCT, utilizing the RATIO model. Consistent with the widely recognized characterization of gut dysbiosis patterns following HSCT [65], our study provides evidence of a comparable trend

in the salivary microbiome. By employing 16S rRNA gene sequencing and analyzing a substantial dataset of 1291 saliva and 504 stool samples, we observed a similar decrease in alpha diversity and an increase in the occurrence of bacterial domination (> 30% relative abundance) primarily attributed to the genera *Veillonella* and *Streptococcus* for the saliva and the genus *Enterococcus* and unclassified members of Enterobacteriaceae for the stool samples. These findings suggest an altered microbial ecosystem post-HSCT in both the oral cavity and the gastrointestinal tract, potentially associated with HSCT complications and outcomes like GVHD, relapse, and overall survival [66–68].

While numerous studies have explored the relationship between the microbiome and various post-HSCT complications, such as associations with alpha diversity, specific taxonomic changes, and prediction of GVHD severity and its binary occurrence [69–74], no previous models have focused on predicting the TTE based on the stool or saliva microbial population. Moreover, previous models have predominantly relied on clinical features to predict the occurrence or survival function of GVHD [38, 41–44, 75, 76]. Therefore, the main objective of this study was to determine whether the composition of the microbiome can serve as an independent predictor of the TTE for seven HSCT complications and outcomes, including AGVHD (II–IV), AGVHD (III–IV), CGVHD, oral CGVHD, relapse, NRM, and OS. By utilizing the RATIO model, a specifically tailored approach designed to accommodate the distinct characteristics of HSCT cohorts characterized by competing events, we effectively enhance the accuracy of traditional survival analysis predictions for both the stool samples (with an average increase of 0.1367 in concordance index, CI) and the saliva samples (with an average increase of 0.091 in CI). This improvement in accuracy is achieved by employing the microbiome alone or in combination with clinical features.

To enhance the practicality and precision of the RATIO model, we conducted a comprehensive analysis aimed at identifying the most critical time points for accurately assessing the risk of each complication. These findings have significant implications for optimizing the application of the RATIO model in clinical settings, enabling clinicians to focus on specific time intervals that yield the most valuable insights and facilitating early intervention that can mitigate potential complications. More specifically, these identified critical time points should be considered as key intervention opportunities for various microbial treatments, including fecal microbiota transplantation [77], probiotics, and prebiotics. By targeting these specific time windows, interventions can be strategically timed

to maximize their effectiveness and enhance patient outcomes.

Utilizing the RATIO model also enables the identification of specific taxa that are associated with each of the events, as well as the exploration of event clustering based on these taxa. For instance, in the stool samples, *Collinsella* and *Eggerthella* were found to be correlated with a shorter time to occurrence for all seven events. *Collinsella*, a gram-positive obligate anaerobe and the most abundant member of the phylum Actinobacteria in a healthy human gut, has been previously linked to increased production of the proinflammatory cytokine IL-17A and gut dysbiosis [78–80]. *Eggerthella*, a member of the class Coriobacteriia, has recently been associated with hospitalization and increased intestinal permeability [81, 82].

To further investigate bacterial patterns associated with HSCT outcomes, we validated our model on an additional cohort containing stool samples from Masetti et al. study. Ten and nine shared taxa were identified between the two cohorts, associated with shorter time to AGVHD and OS, respectively, with eight taxa overlapping between the two events. *Collinsella* and *Eggerthella* were among the taxa shared across the cohorts. Additional shared taxa, including *Methanobrevibacter*, *Alistipes*, *Bifidobacterium*, *Parabacteroides*, and *Bacteroides*, further support the hypothesis of microbial involvement in HSCT complications. *Methanobrevibacter*, a dominant human-associated methanogen, has been shown to influence gut metabolic pathways, which may indirectly modulate immune responses [83]. *Alistipes*, although less abundant, has demonstrated emerging relevance in inflammatory and immune-mediated conditions [84]. *Actinomyces* species, which were found to correlate with shorter time to AGVHD, are typically associated with opportunistic infections and may contribute to gut dysbiosis under the immunosuppressive conditions following HSCT [85]. Conversely, genera such as *Bifidobacterium*, *Bacteroides*, and *Parabacteroides*, commonly linked to short-chain fatty acid (SCFA) production and gut barrier integrity [86–88], may reflect disruptions in overall microbial communities that exacerbate AGVHD risk.

The identification of *Rothia* and unclassified *Bacteroidales* species as shared taxa associated with prolonged time to AGVHD onset suggests their potential protective roles in mitigating HSCT-related complications. *Rothia* has demonstrated anti-inflammatory properties by inhibiting pathogen-induced pro-inflammatory responses through the suppression of the NF- κ B pathway [89]. This anti-inflammatory activity may contribute to maintaining mucosal integrity and potentially reducing the risk of gastrointestinal complications associated with AGVHD. In addition, some species of *Bacteroidales*, e.g., *Bacteroides*

thetaiotaomicron and *Bacteroides fragilis* exhibit anti-inflammatory characteristics and may foster an anti-inflammatory gut environment, thereby delaying the onset of AGVHD. Finally, *Corynebacterium* was found to be associated with improved overall survival in both cohorts. This genus is a member of the microbiota as well and has also been shown to have anti-inflammatory and protective effects [90, 91].

These findings underscore the RATIO model's ability to efficiently calculate time-to-event while identifying relevant microbial taxa that not only serve as predictive biomarkers but also provide mechanistic insights into the pathophysiology of post-HSCT complications. These insights pave the way for future microbiome-targeted interventions aimed at improving HSCT outcomes.

Interestingly, when examining the clustering of events within the saliva cohort, contrasting patterns were observed between GVHD- and relapse-related events and OS and NRM. These features allow for the identification of potential candidates that can enhance our understanding of the underlying mechanism behind the different events. Furthermore, it opens up avenues for the development of novel treatments and preventive regimens, which may potentially lead to improved patient outcomes.

In summary, by leveraging an extensive and unique study population, encompassing thousands of longitudinal stool and saliva samples obtained from HSCT patients, in conjunction with comprehensive clinical data spanning the first year post-HSCT, the RATIO model achieved enhanced predictive accuracy for the TTE of seven post-HSCT events compared to existing survival models.

Conclusions

This comprehensive analysis of stool and saliva microbiomes demonstrates that the RATIO survival analysis model effectively predicts the timing of HSCT complications using microbiome data as a primary input. Notably, this study represents the largest and most detailed investigation of the oral microbiome in the context of HSCT, providing novel insights into its prognostic potential. Even with the inclusion of clinical features, the models maintained robust predictive performance, underscoring the microbiome's role as an independent biomarker. These findings highlight the potential of integrating microbial profiles into clinical decision-making to enhance the precision of interventions and improve patient outcomes. However, translating these results into clinical practice requires prospective, multicenter validation to establish the

model's generalizability and ensure its clinical utility. Such validation will be essential to solidify the role of microbiome-informed strategies in advancing personalized care for HSCT patients.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s13073-025-01507-8>.

Additional File 1

Additional File 2

Additional File 3

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Author contribution

Conceptualization, O.S., A.E., O.K., Y.L., and R.S.; Methodology, O.S., A.E., and R.S.; Validation, O.S.; Formal Analysis, O.S.; Investigation, O.S., A.E., R.S., S.F., M.G., I.D., R.Y., N.S.G., J.A.F., A.S., A.N.; Data Curation, O.S. and A.E.; Writing – Original Draft, O.S., A.E. and R.S.; Visualization, O.S. and A.E.; Supervision, O.K., Y.L. and R.S.; Revision, O.S., A.E., S.T., O.K., Y.L., R.S.; Data of validation cohort, M.F.

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Data availability

The 16S rRNA gene sequence data have been deposited in the European Bioinformatics Institute (EBI) database under accession no. ERP172378 (Stool and Saliva, URL: <https://www.ebi.ac.uk/ena/browser/view/PRJEB89351>). Processed microbiome data is uploaded as Additional File 2 and Additional File 3. The RATIO model is available as a PyPi <https://pypi.org/project/ratio-t2e/> [93].

Declarations

Ethics approval and consent to participate

The study has been approved by the Sheba Medical Center Institutional Review Board (registration number 3822-16-SMC), and all participants provided written informed consent to participate in this study. The study protocol adheres to the ethical guidelines outlined in the 1975 Declaration of Helsinki.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Author details

¹Department of Mathematics, Bar-Ilan University, Ramat Gan 52900, Israel.

²The Azrieli Faculty of Medicine, Bar-Ilan University, Safed, Israel. ³Hematology and Bone Marrow Transplantation Division, Chaim Sheba Medical Center, Tel Hashomer, Ramat Gan, Israel. ⁴Faculty of Medicine, Tel Aviv University, Tel Aviv, Israel. ⁵Weill Cornell Medical College, New York, NY, USA. ⁶Human Microbiomics Unit, Department of Medical and Surgical Sciences, University of Bologna, 40138 Bologna, Italy. ⁷Adult BMT Service, Memorial Sloan Kettering Cancer Center, New York, NY, USA.

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References

- Copelan EA. Hematopoietic stem-cell transplantation. *N Engl J Med*. 2006;354:1813–26.
- Shouval R, Fein JA, Labopin M, Kröger N, Duarte RF, Bader P, et al. Outcomes of allogeneic haematopoietic stem cell transplantation from HLA-matched and alternative donors: a European Society for Blood and Marrow Transplantation registry retrospective analysis. *The Lancet Haematology*. 2019;6:e573–84.
- Wingard JR, Hsu J, Hiemenz JW. Hematopoietic Stem Cell Transplantation: An Overview of Infection Risks and Epidemiology. *Infect Dis Clin North Am*. 2010;24:257–72.
- Ferrara JL, Levine JE, Reddy P, Holler E. Graft-versus-host disease. *Lancet*. 2009;373:1550–61.
- Styczyński J, Tridello G, Iacobelli S, van Biezen A, van der Werf S, et al. Death after hematopoietic stem cell transplantation: changes over calendar year time, infections and associated factors. *Bone Marrow Transplant*. 2020;55:126–36.
- Neuman H, Debelius JW, Knight R, Koren O. Microbial endocrinology: the interplay between the microbiota and the endocrine system. *FEMS Microbiol Rev*. 2015;39:509–21. 2015/02/24 ed.
- Moadi L, Turjeman S, Asulin N, Koren O. The effect of testosterone on the gut microbiome in mice. *Commun Biol*. 2024;7:880.
- Pitlik SD, Koren O. How holobionts get sick—toward a unifying scheme of disease. *Microbiome*. 2017;5:5.
- Glinert A, Turjeman S, Elliott E, Koren O. Microbes, metabolites and (synaptic) malleability, oh my! The effect of the microbiome on synaptic plasticity. *Biol Rev Camb Philos Soc*. 2022;97:582–99.
- Fontaine F, Turjeman S, Callens K, Koren O. The intersection of undernutrition, microbiome, and child development in the first years of life. *Nat Commun*. 2023;14:3554.
- Shouval R, Eshel A, Dubovski B, Kuperman AA, Danylesko I, Fein JA, et al. Patterns of salivary microbiota injury and oral mucositis in recipients of allogeneic hematopoietic stem cell transplantation. *Blood Adv*. 2020;4:2912–7. 2020/07/01 ed.
- Kumari R, Palaniyandi S, Hildebrandt GC. Microbiome: An Emerging New Frontier in Graft-Versus-Host Disease. *Dig Dis Sci*. 2019;64:669–77.
- Ciernikova S, Kasperova B, Drgona L, Smolkova B, Stevurkova V, Mego M. Targeting the gut microbiome: An emerging trend in hematopoietic stem cell transplantation. *Blood Rev*. 2021;48: 100790.
- Shouval R, Waters NR, Gomes ALC, Zuanelli Brambilla C, Fei T, Devlin SM, et al. Conditioning Regimens are Associated with Distinct Patterns of Microbiota Injury in Allogeneic Hematopoietic Cell Transplantation. *Clin Cancer Res*. 2023;29:165–73.
- Nguyen CL, Markey KA, Miltiadous O, Dai A, Waters N, Sadeghi K, et al. High-resolution analyses of associations between medications, microbiome, and mortality in cancer patients. *Cell*. 2023;186:2705–2718.e17.
- Taur Y, Jenq RR, Perales MA, Littmann ER, Morjaria S, Ling L, et al. The effects of intestinal tract bacterial diversity on mortality following allogeneic hematopoietic stem cell transplantation. *Blood*. 2014;124:1174–82. 2014/06/19 ed.
- Andermann TM, Peled JU, Ho C, Reddy P, Riches M, Storb R, et al. The Microbiome and Hematopoietic Cell Transplantation: Past, Present, and Future. *Biol Blood Marrow Transplant*. 2018;24:1322–40.
- Zinter MS, Lindemans CA, Versluys BA, Mayday MY, Sunshine S, Reyes G, et al. The pulmonary metatranscriptome prior to pediatric HCT identifies post-HCT lung injury. *Blood*. 2021;137:1679–89.
- Malard F, Lavelle A, Battipaglia G, Gaugler B, Dulery R, Brissot E, et al. Impact of gut fungal and bacterial communities on the outcome of allogeneic hematopoietic cell transplantation. *Mucosal Immunol*. 2021;14:1127–32.
- Zinter MS, Versluys AB, Lindemans CA, Mayday MY, Reyes G, Sunshine S, et al. Pulmonary microbiome and gene expression signatures differentiate lung function in pediatric hematopoietic cell transplant candidates. *Sci Transl Med*. 2022;14: eabm8646.
- Jenq RR, Taur Y, Devlin SM, Ponce DM, Goldberg JD, Ahr KF, et al. Intestinal Blautia Is associated with reduced death from graft-versus-host disease. *Biol Blood Marrow Transplant*. 2015;21:1373–83. 2015/05/16 ed.
- Hong T, Wang R, Wang X, Yang S, Wang W, Gao Q, et al. Interplay Between the Intestinal Microbiota and Acute Graft-Versus-Host Disease: Experimental Evidence and Clinical Significance. *Front Immunol*. 2021;12: 644982.
- Stein-Thoeringer CK, Nichols KB, Lazrak A, Docampo MD, Slingerland AE, Slingerland JB, et al. Lactose drives Enterococcus expansion to promote graft-versus-host disease. *Science*. 2019;366:1143–9. 2019/11/30 ed.
- Peled JU, Devlin SM, Staffas A, Lumish M, Khanin R, Littmann ER, et al. Intestinal microbiota and relapse after hematopoietic-cell transplantation. *J Clin Oncol*. 2017;35:1650–9. 2017/03/16 ed.
- Yamamoto A, Kambara Y, Fujiwara H. Impact of oral microbiota on pathophysiology of GVHD. *Front Immunol*. 2023;14: 1132983.
- Holtan SG, Verneris MR, Schultz KR, Newell LF, Meyers G, He F, et al. Circulating angiogenic factors associated with response and survival in patients with acute graft-versus-host disease: results from Blood and Marrow Transplant Clinical Trials Network 0302 and 0802. *Biol Blood Marrow Transplant*. 2015;21:1029–36.
- Levine JE, Logan BR, Wu J, Alousi AM, Bolaños-Meade J, Ferrara JLM, et al. Acute graft-versus-host disease biomarkers measured during therapy can predict treatment outcomes: a Blood and Marrow Transplant Clinical Trials Network study. *Blood*. 2012;119:3854–60.
- Urbano-Ispizua A, Rozman C, Pimentel P, Solano C, de la Rubia J, Brunet S, et al. Risk factors for acute graft-versus-host disease in patients undergoing transplantation with CD34+ selected blood cells from HLA-identical siblings. *Blood*. 2002;100:724–7.
- Remberger M, Persson U, Hauzenberger D, Ringdén O. An association between human leucocyte antigen alleles and acute and chronic graft-versus-host disease after allogeneic haematopoietic stem cell transplantation. *Br J Haematol*. 2002;119:751–9.
- Chen Y, Li G, Jiang W, Nie RC, Deng H, Chen Y, et al. Prognostic risk factor of major salivary gland carcinomas and survival prediction model based on random survival forests. *Cancer Med*. 2023;12:10899–907.
- Andrade C. Survival Analysis, Kaplan-Meier Curves, and Cox Regression: Basic Concepts. *Indian J Psychol Med*. 2023;45:434–5.
- Holtan SG, DeFor TE, Lazaryan A, Bejanyan N, Arora M, Brunstein CG, et al. Composite end point of graft-versus-host disease-free, relapse-free survival after allogeneic hematopoietic cell transplantation. *Blood*. 2015;125:1333–8.
- Sorror ML, Martin PJ, Storb RF, Bhatia S, Maziari RT, Pulsipher MA, et al. Pretransplant comorbidities predict severity of acute graft-versus-host disease and subsequent mortality. *Blood*. 2014;124:287–95.
- Levine JE, Logan B, Wu J, Alousi AM, Ho V, Bolaños-Meade J, et al. Graft-versus-host disease treatment: predictors of survival. *Biol Blood Marrow Transplant*. 2010;16:1693–9.
- Baron F, Maris MB, Sandmaier BM, Storer BE, Sorror M, DiConcescu R, et al. Graft-versus-tumor effects after allogeneic hematopoietic cell transplantation with nonmyeloablative conditioning. *J Clin Oncol*. 2005;23:1993–2003.
- Pintilie M. Competing risks: a practical perspective. Ontario Cancer Institute Canada: Wiley;
- Fine JP, Jiang H, Chappell R. On Semi-Competing Risks Data. *Biometrika*. 2001;88:907–19.
- Lee C, Lee SJ, Haneuse S. Time-to-event analysis when the event is defined on a finite time interval. *Stat Methods Med Res*. 2020;29:1573–91.
- Hsieh J-J, Huang Y-T. Regression analysis based on conditional likelihood approach under semi-competing risks data. *Lifetime Data Anal*. 2012;18:302–20.
- Lakhal L, Rivest L-P, Abdous B. Estimating survival and association in a semicompeting risks model. *Biometrics*. 2008;64:180–8.
- Tchetgen Tchetgen EJ. Identification and estimation of survivor average causal effects. *Stat Med*. 2014;33:3601–28.
- Egleston BL, Scharfstein DO, Freeman EE, West SK. Causal inference for non-mortality outcomes in the presence of death. *Biostatistics*. 2007;8:526–45.
- Xu J, Kalbfleisch JD, Tai B. Statistical analysis of illness-death processes and semicompeting risks data. *Biometrics*. 2010;66:716–25.
- Lee KH, Haneuse S, Schrag D, Dominici F. Bayesian Semi-parametric Analysis of Semi-competing Risks Data: Investigating Hospital

- Readmission after a Pancreatic Cancer Diagnosis. *J R Stat Soc Ser C Appl Stat.* 2015;64:253–73.
45. Shtossel O, Dubovski B, Koren O, Louzoun Y. Left barrier loss for unbiased survival analysis prediction. 2022. Available from: <https://www.researchsquare.com/article/rs-1945158/v1>. Cited 2025 Jan 13.
 46. Harris PA, Taylor R, Thielke R, Payne J, Gonzalez N, Conde JG. Research electronic data capture (REDCap)—a metadata-driven methodology and workflow process for providing translational research informatics support. *J Biomed Inform.* 2009;42:377–81.
 47. Rowlings PA, Przepiorka D, Klein JP, Gale RP, Passweg JR, Henslee-Downey PJ, et al. IBMTR Severity Index for grading acute graft-versus-host disease: retrospective comparison with Glucksberg grade. *Br J Haematol.* 1997;97:855–64.
 48. Jagasia MH, Greinix HT, Arora M, Williams KM, Wolff D, Cowen EW, et al. National Institutes of Health Consensus Development Project on Criteria for Clinical Trials in Chronic Graft-versus-Host Disease: I. The 2014 Diagnosis and Staging Working Group report. *Biol Blood Marrow Transplant.* 2015;21:389–401.e1.
 49. Youngster I, Eshel A, Geva M, Danylesko I, Henig I, Zuckerman T, et al. Fecal microbiota transplantation in capsules for the treatment of steroid refractory and steroid dependent acute graft vs. host disease: a pilot study. *Bone Marrow Transplant.* 2024;59:409–16.
 50. Eshel A, Sharon I, Nagler A, Bomze D, Danylesko I, Fein JA, et al. Blood-stream infections' origins following fecal microbiota transplantation: a strain-level analysis. *Blood Adv.* 2021/10/14 ed. 2021; Available from: <https://www.ncbi.nlm.nih.gov/pubmed/34644375>.
 51. Bolyen E, Rideout JR, Dillon MR, Bokulich NA, Abnet CC, Al-Ghalith GA, et al. Reproducible, interactive, scalable and extensible microbiome data science using QIIME 2. *Nat Biotechnol.* 2019/07/26 ed. 2019;37:852–7.
 52. Callahan BJ, McMurdie PJ, Rosen MJ, Han AW, Johnson AJ, Holmes SP. DADA2: High-resolution sample inference from Illumina amplicon data. *Nat Methods.* 2016/05/24 ed. 2016;13:581–3.
 53. DeSantis TZ, Hugenholtz P, Larsen N, Rojas M, Brodie EL, Keller K, et al. Greengenes, a chimera-checked 16S rRNA gene database and workbench compatible with ARB. *Appl Environ Microbiol.* 2006;72:5069–72.
 54. Jasner Y, Belogolovski A, Ben-Itzhak M, Koren O, Louzoun Y. Microbiome preprocessing machine learning pipeline. *Front Immunol.* 2021;12: 677870.
 55. Cox DR. Regression models and life-tables. *J R Stat Soc Ser B Stat Methodol.* 1972;34:187–202.
 56. Ishwaran H, Kogalur UB, Blackstone EH, Lauer MS. Random survival forests. *Ann Appl Stat.* 2008;2. Available from: <https://projecteuclid.org/journals/annals-of-applied-statistics/volume-2/issue-3/Random-survival-forests/https://doi.org/10.1214/08AOAS169>. Cited 2025 Jan 13.
 57. Wang H, Li G. A selective review on random survival forests for high dimensional data. *Quant Biosci.* 2017;36:85–96.
 58. Gerds TA (2025). prodlim: product-limit estimation for censored event history analysis. R package version 2025.04.28, <https://github.com/tagteam/prodlim>.
 59. Mogensen UB, Ishwaran H, Gerds TA. Evaluating random forests for survival analysis using prediction error curves. *J Stat Softw.* 2012;50:1–23.
 60. Salerno S, Li Y. High-dimensional survival analysis: methods and applications. *Annu Rev Stat Appl.* 2023;10:25–49.
 61. Henderson R. Problems and prediction in survival-data analysis. *Stat Med.* 1995;14:161–84.
 62. Sinharay S. Jackknife Methods. In: Peterson P, Baker E, McGaw B, editors. *International Encyclopedia of Education* (Third Edition). Oxford: Elsevier; 2010. p. 229–31. Available from: <https://www.sciencedirect.com/science/article/pii/B9780080448947013385>.
 63. Lundberg S, Lee S-I. A Unified Approach to Interpreting Model Predictions. *arXiv*; 2017. Available from: <https://arxiv.org/abs/1705.07874>. Cited 2025 Jan 13.
 64. Masetti R, Leardini D, Muratore E, Fabbriini M, D'Amico F, Zama D, et al. Gut microbiota diversity before allogeneic hematopoietic stem cell transplantation as a predictor of mortality in children. *Blood.* 2023;142:1387–98.
 65. Shono Y, van den Brink MRM. Gut microbiota injury in allogeneic haematopoietic stem cell transplantation. *Nat Rev Cancer.* 2018;18:283–95.
 66. Heidrich V, Knebel FH, Bruno JS, de Molla VC, Miranda-Silva W, Asprino PF, et al. Longitudinal analysis at three oral sites links oral microbiota to clinical outcomes in allogeneic hematopoietic stem-cell transplant. *Microbiol Spectr.* 2023;11: e0291023.
 67. de Molla VC, Heidrich V, Bruno JS, Knebel FH, Miranda-Silva W, Asprino PF, et al. Disruption of the oral microbiota is associated with a higher risk of relapse after allogeneic hematopoietic stem cell transplantation. *Sci Rep.* 2021;11:17552.
 68. Burgos da Silva M, Ponce DM, Dai A, M Devlin S, Gomes ALC, Moore G, et al. Preservation of the fecal microbiome is associated with reduced severity of graft-versus-host disease. *Blood.* 2022;140:2385–97.
 69. Henig I, Yehudai-Ofir D, Zuckerman T. The clinical role of the gut microbiome and fecal microbiota transplantation in allogeneic stem cell transplantation. *haematol.* 2020; Available from: <https://haematologica.org/article/view/haematol.2020.247395>. Cited 2025 Jan 13.
 70. Han L, Zhang H, Chen S, Zhou L, Li Y, Zhao K, et al. Intestinal microbiota can predict acute graft-versus-host disease following allogeneic hematopoietic stem cell transplantation. *Biol Blood Marrow Transplant.* 2019;25:1944–55.
 71. Fredricks DN. The gut microbiota and graft-versus-host disease. *J Clin Invest.* 2019;129:1808–17.
 72. Sofi MH, Wu Y, Ticer T, Schutt S, Bastian D, Choi H-J, et al. A single strain of *Bacteroides fragilis* protects gut integrity and reduces GVHD. *JCI Insight.* 2021;6: 136841.
 73. Laheij AMGA, Rozema FR, Brennan MT, von Bültzingslöwen I, van Leeuwen SJM, Potting C, et al. Long-Term Analysis of Resilience of the Oral Microbiome in Allogeneic Stem Cell Transplant Recipients. *Microorganisms.* 2022;10:10.
 74. Microbiota long-term dynamics and prediction of acute graft-versus-host disease in pediatric allogeneic stem cell transplantation. *England*; 2021
 75. Arai Y, Kondo T, Fuse K, Shibasaki Y, Masuko M, Sugita J, et al. Using a machine learning algorithm to predict acute graft-versus-host disease following allogeneic transplantation. *Blood Adv.* 2019;3:3626–34.
 76. Shouval R, Labopin M, Bondi O, Mishan-Shamay H, Shimoni A, Ciceri F, et al. Prediction of Allogeneic Hematopoietic Stem-Cell Transplantation Mortality 100 Days After Transplantation Using a Machine Learning Algorithm: A European Group for Blood and Marrow Transplantation Acute Leukemia Working Party Retrospective Data Mining Study. *J Clin Oncol.* 2015;33:3144–51.
 77. Shtossel O, Turjeman S, Riumin A, Goldberg MR, Elizur A, Bekor Y, et al. Recipient-independent, high-accuracy FMT-response prediction and optimization in mice and humans. *Microbiome.* 2023;11:181.
 78. Chen J, Wright K, Davis JM, Jeraldo P, Marietta EV, Murray J, et al. An expansion of rare lineage intestinal microbes characterizes rheumatoid arthritis. *Genome Med.* 2016;8:43.
 79. Astbury S, Atallah E, Vijay A, Aithal GP, Grove JJ, Valdes AM. Lower gut microbiome diversity and higher abundance of proinflammatory genus *Collinsella* are associated with biopsy-proven nonalcoholic steatohepatitis. *Gut Microbes.* 2020;11:569–80.
 80. Kwon J, Bae M, Szamosvári D, Cassilly CD, Bolze AS, Jackson DR, et al. *Collinsella aerofaciens* Produces a pH-Responsive Lipid Immunogen. *J Am Chem Soc.* 2023;145:7071–4.
 81. Mizutani T, Ishizaka A, Koga M, Ikeuchi K, Saito M, Adachi E, et al. Correlation analysis between gut microbiota alterations and the cytokine response in patients with coronavirus disease during hospitalization. *Microbiol Spectr.* 2022;10: e0168921.
 82. Balakrishnan B, Luckey D, Taneja V. Autoimmunity-Associated Gut Commensals Modulate Gut Permeability and Immunity in Humanized Mice. *Mil Med.* 2019;184:529–36.
 83. Djemai K, Drancourt M, Tidjani AM. Bacteria and methanogens in the human microbiome: a review of syntrophic interactions. *Microb Ecol.* 2022;83:536–54.
 84. Parker BJ, Wearsch PA, Veloo ACM, Rodriguez-Palacios A. The Genus *Alistipes*: gut bacteria with emerging implications to inflammation, cancer, and mental health. *Front Immunol.* 2020;11: 906.
 85. Könönen E, Wade WG. Actinomyces and related organisms in human infections. *Clin Microbiol Rev.* 2015;28:419–42.
 86. Cui Y, Zhang L, Wang X, Yi Y, Shan Y, Liu B, et al. Roles of intestinal *Parabacteroides* in human health and diseases. *FEMS Microbiology Letters.* 2022;369: fnac072.
 87. Fusco W, Lorenzo MB, Cintoni M, Porcari S, Rinninella E, Kaitsas F, et al. Short-chain fatty-acid-producing bacteria: key components of the human gut microbiota. *Nutrients.* 2023;15: 2211.

88. Cheng J, Hu J, Geng F, Nie S. Bacteroides utilization for dietary polysaccharides and their beneficial effects on gut health. *Food Sci Human Wellness*. 2022;11:1101–10.
89. Rigauts C, Aizawa J, Taylor SL, Rogers GB, Govaerts M, Cos P, et al. *Rothia mucilaginosa* is an anti-inflammatory bacterium in the respiratory tract of patients with chronic lung disease. *Eur Respir J*. 2022;59:2101293.
90. Horn KJ, Jaber Vivar AC, Arenas V, Andani S, Janoff EN, Clark SE. *Corynebacterium* Species Inhibit *Streptococcus pneumoniae* Colonization and Infection of the Mouse Airway. *Front Microbiol*. 2022;12: 804935.
91. Menberu MA, Liu S, Cooksley C, Hayes AJ, Psaltis AJ, Wormald P-J, et al. *Corynebacterium accolens* Has Antimicrobial Activity against *Staphylococcus aureus* and Methicillin-Resistant *S. aureus* Pathogens Isolated from the Sinonasal Niche of Chronic Rhinosinusitis Patients. *Pathogens*. 2021;10: 207.
92. Calle ML, Pujolassos M, Susin A. coda4microbiome: compositional data analysis for microbiome cross-sectional and longitudinal studies. *BMC Bioinformatics*. 2023;24(1):82.
93. Shtossel O, Koren O, Louzoun Y. PyPi for Left barrier loss for unbiased survival analysis prediction, <https://pypi.org/project/ratio-t2e/>.
94. Spyridonidis A, et al. Redefining and measuring transplant conditioning intensity in current era: a study in acute myeloid leukemia patients. *Bone Marrow Transplantation*. 2020;55(6):1114–25.

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